

Logistic Regression Integrated Autoencoder Framework for Intelligent Acoustic Anomaly Detection in Industrial Environments

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Abstract—Early detection of anomalous sound patterns in industrial environments is essential for preventing equipment failures and ensuring operational continuity. However, automated frameworks for acoustic anomaly detection remain challenging to develop due to the inherent variability of anomalous signals and the lack of a standardized definition of “abnormality.” Recent advances in deep learning have enabled the design of intelligent diagnostic systems capable of modeling complex time-series patterns. In this study, an autoencoder-based deep learning framework coupled with a logistic regression classifier is proposed for the detection of abnormal acoustic events in manufacturing settings. The autoencoder efficiently captures the latent representation of normal operational sound, while logistic regression discriminates between regular and anomalous patterns based on reconstruction error features. Experimental evaluations on real-world industrial sound datasets demonstrate that the proposed hybrid model achieves an anomaly detection accuracy of 97%, confirming its robustness and practical potential for real-time fault detection and predictive maintenance in industrial systems.

Index Terms—Acoustic anomaly detection, Autoencoder, Logistic regression, Intelligent monitoring system, Industrial sound analysis

I. INTRODUCTION

Sound anomaly detection is a critical area in machine learning and signal processing that focuses on identifying unusual or abnormal acoustic events within audio streams. These anomalies may indicate malfunctions, security breaches, or hazardous situations, making early detection essential across diverse domains such as industrial monitoring, healthcare, en-

vironmental surveillance, and smart home systems. Although considerable progress has been made in pattern recognition, constructing automated frameworks for sound anomaly detection continues to pose substantial challenges. These difficulties primarily stem from the intrinsic variability of anomalous acoustic signals and the lack of a universally accepted definition of what constitutes “abnormal” sound. In recent years, however, the advent of deep learning techniques has significantly propelled the development of advanced, data-driven approaches for identifying acoustic anomalies with greater accuracy and adaptability.

Sound anomaly detection has garnered increasing attention over the past decade, driven by its wide range of applications in industrial monitoring, environmental sensing, healthcare, and smart infrastructure. Early approaches relied heavily on statistical models and signal processing techniques, such as Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs), which required extensive feature engineering and struggled with generalizing to complex acoustic environments. The introduction of machine learning, particularly deep learning, has significantly transformed this field by enabling the automatic extraction of high-level features from raw audio data. Techniques such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and more recently, transformer-based models have demonstrated promising performance in modeling temporal and spectral characteristics of sound. In parallel, unsupervised and self-supervised learning paradigms have gained prominence due to the scarcity of

labeled anomalous data, leading to the development of robust systems trained primarily on normal sound patterns. This evolution reflects a growing shift from rule-based detection to data-driven intelligence in addressing the complex and dynamic nature of acoustic anomalies. Notably, autoencoder architectures, which is widely recognized for their capacity to learn compressed representations, have shown significant promise in detecting irregular patterns in time-series data. Particularly within industrial environments, autoencoder-based frameworks trained on sound samples have demonstrated potential for identifying device malfunctions through the recognition of anomalous acoustic signatures. This study explores the integration of autoencoder models with robust anomaly detection strategies to predict anomalous sound events indicative of equipment failure in industrial settings. While the combination of autoencoders and thresholding techniques has yielded encouraging results, the task remains complex. Conventional thresholding approaches often struggle to accurately distinguish between normal and anomalous acoustic signals. In light of these challenges, this research aims to establish a more effective framework for industrial sound anomaly detection by addressing the limitations inherent in existing autoencoder-based methods. The contributory objectives of the proposed work are outlined below.

- **Development of an LSTM-autoencoder Model:**

An LSTM autoencoder architecture is trained exclusively with normal sound samples so that it can reconstruct them with smaller error as compared to the anomalous sound samples.

- **Development of a Logistic Regression Model:**

The trained LSTM autoencoder is fed with both both anomalous and normal instances of sound samples and a set of error vector is prepared by making comparison between the input and the output vector. Each error vector is labeled with either anomalous or normal class depending on the category of its source. These error vectors are used in developing a audio anomaly discriminatory model using logistic regression strategy.

The logistic regression model is trained with the set of labeled error vectors to classify them into anomalous or normal instances. The novelty of the proposed work lies in utilizing the logistic regression model instead of popular thresholding techniques for discriminating normal sound patterns from the anomalous patterns.

- **Deployment of the Proposed Framework integrating trained LSTMautoencoder and Logistic Regression Model:**

During deployment of the proposed framework, an audio signal is fed to the trained LSTM autoencoder and the corresponding error vector is generated by computing deviation of the output vector from the input vector. Subsequently, the trained logistic regression model is applied to the error vector for detecting its class (normal or anomalous) which also represents the category of the input audio signal.

II. RELATED WORK

Sound anomaly detection has evolved significantly over the past two decades, driven by the growing demand for intelligent monitoring systems across industrial, environmental, and urban contexts. Traditional approaches primarily relied on statistical models and feature engineering techniques. For instance, Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs) were among the earliest probabilistic methods applied to acoustic anomaly detection [1], [2]. These models required domain-specific handcrafted features such as Mel-Frequency Cepstral Coefficients (MFCCs), spectral roll-off, and zero-crossing rate, and were limited in their ability to model the complex temporal and spectral characteristics of real-world audio.

With the rise of deep learning, a paradigm shift has occurred toward end-to-end models capable of learning hierarchical representations directly from raw or minimally processed audio. Autoencoders (AEs), in particular, have emerged as a dominant approach for unsupervised anomaly detection, wherein a model is trained to reconstruct normal sound data and high reconstruction errors are interpreted as anomalies [3], [4]. Variants such as Convolutional Autoencoders (CAEs) and Recurrent Autoencoders (RAEs) have been employed to improve feature extraction and capture temporal dependencies in sound signals [5].

In parallel, the use of Generative Adversarial Networks (GANs) has also been explored. [6] introduced WaveGAN, which synthesized audio waveforms, and subsequent work extended GANs for anomaly detection by comparing input signals with GAN-generated outputs to identify discrepancies [7]. However, training GANs remains challenging due to instability and mode collapse, particularly when data is scarce or imbalanced.

Self-supervised learning has gained prominence in recent years for addressing the scarcity of labeled anomalous data. Models such as CPC (Contrastive Predictive Coding) and wav2vec leverage pretext tasks to learn representations that are useful for downstream anomaly detection [8], [9]. These approaches allow models to generalize better across domains and reduce reliance on large labeled datasets.

Transformer-based models, such as AST (Audio Spectrogram Transformer) and PaSST (Patchout Spectrogram Transformer), have recently demonstrated state-of-the-art performance in a variety of audio classification tasks, including anomaly detection [10], [11]. These models utilize self-attention mechanisms to capture long-range dependencies in audio spectrograms and have outperformed CNNs in terms of robustness and generalization.

Benchmark datasets and evaluation protocols have also played a crucial role in advancing the field. The DCASE (Detection and Classification of Acoustic Scenes and Events) Challenge, particularly Task 2, has provided a standardized framework for evaluating unsupervised anomaly detection systems in machine operating sounds [5]. These benchmarks have encouraged the development of more reproducible and

comparable research, with a focus on real-world applicability in industrial environments.

Despite these advances, several challenges remain. Many deep learning models suffer from high computational costs, making them difficult to deploy on edge devices. Moreover, achieving low false positive rates while maintaining high sensitivity remains an open problem, especially in acoustically diverse or non-stationary environments. Researchers are increasingly exploring hybrid models, combining unsupervised learning with domain adaptation, semi-supervised learning, and attention mechanisms to enhance model robustness and interpretability [12], [13].

In summary, the literature indicates a rapid evolution from rule-based and statistical methods to deep learning-based frameworks for sound anomaly detection. While significant progress has been made, especially in representation learning and model scalability, continued innovation is needed to address the complexities of real-world deployment, particularly in industrial and resource-constrained settings.

III. PROPOSED METHODOLOGY

The development of the proposed framework primarily consists of three phases: 1) Data Preprocessing Phase, 2) Autoencoder Model Training Phase, 3) Logistic Regression Model Training Phase as shown in Fig. 1. During the data preprocessing phase, an audio signal is transformed into short-time energy (STE), short-time Fourier transform (STFT), and Mel frequency cepstral coefficients (MFCC) based feature vectors. In the proposed framework, an LSTM-autoencoder architecture is exclusively trained with these three types of feature vectors obtained from normal/non-anomalous audio signals while aiming for reconstructing them as its output with minimal error. The trained LSTM-autoencoder model is employed to generate a set of error vectors by computing the deviations of reconstructed outputs for both normal and anomalous audio signals given as input. These error vectors are used to train a logistic regression model for classifying them into the normal or anomalous classes depending on the categories of their respective audio signal sources.

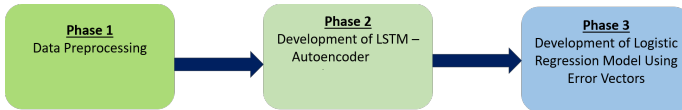


Fig. 1. Overall Flow of the Proposed Framework Development

A. Data Acquisition

The experimental analysis in this study utilizes the "Anomaly Detection from Sound Data Fan" dataset, which is derived from a subset of the MIMII Dataset (Malfunctioning Industrial Machine Investigation and Inspection) [14]. The MIMII dataset was specifically designed to support research in acoustic-based fault diagnosis in industrial equipment. The subset employed in this work focuses on machine sounds

generated by industrial fans operating under both normal and faulty conditions.

The dataset comprises monaural audio recordings sampled at 16 kHz, with each file having a duration of 10 seconds. These recordings were captured in controlled environments using consistent microphone setups to ensure uniform acoustic quality. The dataset includes two primary classes: normal operational sounds and anomalous sounds that simulate various types of mechanical malfunctions (e.g., bearing wear or blade defects).

A total of 9,738 audio samples were used in this study, segmented into three disjoint subsets: 6,521 recordings for model training, 1,875 for validation, and 1,342 for testing. Notably, only normal data were used during the training phase, aligning with the unsupervised anomaly detection paradigm, where the system is trained without prior exposure to anomalous data. The evaluation is then conducted on a mixed test set containing both normal and anomalous instances.

This dataset provides a realistic and challenging benchmark for evaluating the effectiveness of automated anomaly detection systems in the context of industrial condition monitoring, particularly for predictive maintenance of rotating machinery such as fans.

B. Data Pre-processing

During the data preprocessing phase, an audio signal is transformed into short-time energy (STE), short-time Fourier transform (STFT), and Mel frequency cepstral coefficients (MFCC) based feature vectors.

Short time energy (STE) based Features: The sound signals are non-stationary in nature, and they can be transformed into a sequence of small portions of quasistationary signals using the framing/windowing method. The short-time energy of a signal within each frame can be used to form a feature vector for developing the machine learning models.

STFT based Features: The Short-Time Fourier Transform (STFT) is a widely used method for analyzing the frequency content of audio signals over time. Unlike the standard Fourier Transform, which provides frequency information over an entire signal, the STFT divides a signal into overlapping short segments and applies the Fourier Transform to each segment individually. In the proposed work, Linear Predictive Coding (LPC) coefficients from each signal segment were derived after applying STFT for generating Spectral Envelope-Based audio features.

Mel Frequency Cepstral Coefficients (MFCC) based Features: An audio clip's cepstral representation serves as the basis for MFCCs. MFCCs represent the short-time power spectrum of an audio clip based on the discrete cosine transform of the log power spectrum on a nonlinear mel scale. The evenly distributed frequency bands in MFCCs closely resemble the human auditory system, establishing them as an essential component in numerous audio signal processing applications.

C. Autoencoder Development Phase

An autoencoder model comprises of three main components: the encoder block, hidden layer, and decoder block, as depicted in Fig. 2. In our proposed work, it is trained solely by using non-anomalous audio samples. Usually in autoencoder architecture, an input vector is compressed and encoded into the hidden layer by the encoder block, which is then used by the decoder block for reconstruction of input as output. During training, the loss function is defined based on the reconstruction error, which measured in terms of the difference between the input vector and the output vector produced by the model.

The proposed framework employs LSTM (Long Short-Term Memory) autoencoder architecture which is a special variant of Recurrent Neural Networks tailored to effectively model temporal sequences and long-range dependencies with greater precision compared to conventional RNNs. Both encoder and decoder blocks of our LSTM autoencoder architecture consist of three neural layers having 256, 128 and 64 neurons respectively. The structure of the architecture is illustrated in Fig. 2. At each neural layer, ReLU activation function with a dropout rate of 0.5 is applied. The models were trained for 100 epochs with a learning rate of 0.001 using ADAM as optimizer and Mean Absolute Error (MAE) as loss function. During training of the LSTM-autoencoder model, 80% of the normal samples were utilized for training, while the remaining data were reserved for validation. Essentially, the training process involves iterative adjustments of neural link weights in both the encoder and decoder to minimize the loss function.

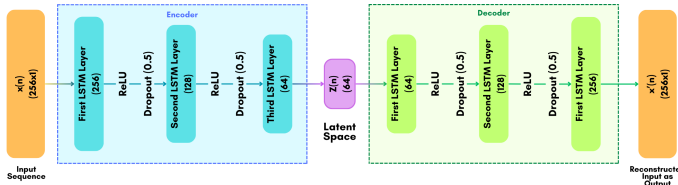


Fig. 2. LSTM Autoencoder Architecture in the Proposed Framework

D. Discriminator Model Development Phase

The objective of the discriminator model development phase is to develop a logistic regression model using error vectors constructed by making MAE based comparison between the input sequence vectors and the output vectors reconstructed by the autoencoder model for both normal and anomalous audio signals. Our logistic regression model adopts a simple classification strategy as it decides the class of an instance based on optimizing a Log Loss cost function (Eqn. 3) employed for improving the accuracy of the estimated probability obtained from sigmoid function (Eqn. 2). Fig. 3 presents a simple architecture of the logistic regression model employed in the proposed work. In order to develop the logistic regression model, 80% error vectors were used as trainee set while rest

of the error vectors were used as testing set.

$$z = w^T x + b \quad (1)$$

$$\hat{y} = P(y = 1|x) = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (2)$$

$$L(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (3)$$

where

- The input error vector, weight vector, and the bias term are denoted by x , w and b respectively.
- $\sigma(z)$ represents the sigmoid activation function.
- L is the log loss function, y is the true label, and $\hat{y}$ is the predicted probability for anomalous instance.

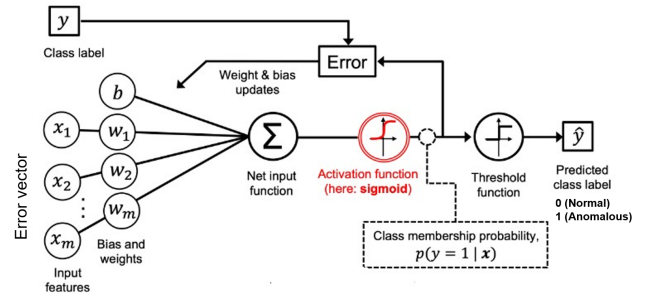


Fig. 3. Schematic Diagram of the Logistic Regression Model as Anomalous Sound Classifier

E. Anomaly Detection Strategy

During the deployment phase, the proposed framework initially transformed the audio signal into STE, STFT and MFCC feature vectors. These feature vectors are subsequently input into the trained LSTM-autoencoder model and the reconstructed outputs by the model are used to generate error vectors by measuring the mean absolute error (MAE)-based deviations from their corresponding inputs. Next, the trained logistic regression model, functioning as a binary classifier, processes these error vectors to predict the class of the audio signal categorizing it as either anomalous or normal instance. Our sound anomaly detection strategy employed in the proposed framework is outlined in Algorithm 1.

IV. EXPERIMENTAL EVALUATION

This section presents the experimental evaluation of the proposed framework to assess its performance using a publicly available dataset obtained from the Kaggle repository [15]. The dataset, originally derived from a subset of the MIMII Dataset (Malfunctioning Industrial Machine Investigation and Inspection) [14], comprises acoustic recordings of industrial fan operations captured under controlled conditions. It includes both normal and anomalous sound events, facilitating the development and assessment of machine learning models for sound-based anomaly detection. The dataset contains a total of 9,738 audio recordings, of which 6,521 normal audio samples

Algorithm 1 Algorithm for Detection of Anomalous Sound Signal

Require: Audio Time Series Data as Input**Ensure:** Prediction of Category of Audio Signal (Anomalous or Normal) as Output

- 1: Transform the audio signal into STE, STFT and MFCC based feature vectors, denoted as S_1 , S_2 and S_3 respectively
 - 2: $AnomalousCount \leftarrow 0$
 - 3: $NormalCount \leftarrow 0$
 - 4: **for** $i = 1$ to 3 **do**
 - 5: Feed S_i to the LSTM-Autoencoder trained with normal audio samples and obtain its output R_i as reconstructed vector
 - 6: $E_i \leftarrow MAE(S_i, R_i)$ ▷ Generate Error Vector
 - 7: Feed E_i to the Logistic Regression Model trained with error vectors and obtain its output as the prediction probability $P_i(y_i = 1|E_i)$ for anomalous instance
 - 8: **if** $P_i(y_i = 1|E_i) > 0.5$ **then**
 - 9: $AnomalousCount \leftarrow AnomalousCount + 1$
 - 10: **else**
 - 11: $NormalCount \leftarrow NormalCount + 1$
 - 12: **end if**
 - 13: **end for**
 - 14: **if** $AnomalousCount > NormalCount$ **then**
 - 15: $AudioCategory \leftarrow Anomalous$
 - 16: **else**
 - 17: $AudioCategory \leftarrow Normal$
 - 18: **end if**
-

are utilized for training, 1,875 for validation, and 1,342 for testing. This partitioning ensures an effective balance between model training, hyperparameter tuning, and performance evaluation.

A. Evaluation Metrics

The popular evaluation metrics such as accuracy (Acc), precision (Pre), and recall (Rec) were used to evaluate the performance of our framework compared to other models. In our experimentation, correct prediction of anomalous sound signals are designated as true positive case while their incorrect predictions as normal instances are designated as false negative instances. Similarly, correct predictions of normal sound signals are designated as true negative case while their incorrect predictions as anomalous instances are designated as false positive case. The equations for computing the evaluation

metrics are presented below.

$$Acc = \frac{N_{TP} + N_{TN}}{N_{Test}} \quad (4)$$

$$Pre = \frac{N_{TP}}{N_{TP} + N_{FP}} \quad (5)$$

$$Rec = \frac{N_{TP}}{N_{TP} + N_{FN}} \quad (6)$$

$$F1_{Score} = \frac{2 \times Pre \times Rec}{Pre + Rec} \quad (7)$$

$$(8)$$

where

- N_{TP} denotes number of true positive instances i.e. number of correct prediction of anomalous sound signals
- N_{TN} denotes number of true negative instances i.e. number of correct prediction of normal sound signals
- N_{FP} denotes number of false positive instances i.e. number of normal sound signals falsely predicted as anomalous sound signals
- N_{FN} denotes number of false negative instances i.e. number of anomalous sound signals falsely predicted as normal sound signals
- N_{Test} denotes total number of sound signals used in testing

In the proposed study, accuracy refers to the percentage of correctly identified instances of both anomalous and normal sound signals out of the all testing samples. Precision metric is computed as the proportion of correctly identified positive instances among all positives predicted by the model. The ability to correctly detect all actual positive cases is measured by Recall (also called sensitivity) metric. In general, high accuracy indicates the suitability of a model while high precision ensures fewer false alarms. F1 score considers both precision and recall values in terms of a single evaluation metric for obtaining a better insight while judging the performance of a classification model.

B. Results and Discussion

TABLE I
EXPERIMENTAL RESULTS FOR ANOMALY DETECTION FROM SOUND DATA-FAN

Method	Acc (%)	Pre (%)	Rec (%)	F1 (%)
STE-Autoencoder-Thresholding	84.45	77.95	98.75	81.07
STFT-Autoencoder-Thresholding	88.76	83.14	97.43	85.86
MFCC-Autoencoder-Thresholding	92.56	88.31	98.77	90.39
Ensemble-Autoencoder-Thresholding	96.83	95.94	96.76	96.38
STE-Autoencoder-Logistic	87.85	81.84	97.78	84.74
STFT-Autoencoder-Logistic	91.43	94.23	98.81	92.81
MFCC-Autoencoder-Logistic	94.18	95.31	97.78	94.74
Ensemble-Autoencoder-Logistic	97.07	96.86	98.01	97.36

Table I presents the performance evaluation of the proposed framework in comparison with several widely used baseline

strategies, using standard classification metrics. The experimental results are reported for three types of acoustic features: Short-Time Energy (STE), Short-Time Fourier Transform (STFT), and Mel-Frequency Cepstral Coefficients (MFCC), under two different anomaly detection strategies: thresholding and logistic regression. While the recall values across all evaluated models remain relatively consistent, the proposed ensemble-based approach demonstrates superior performance in terms of both accuracy and precision. This ensemble framework aggregates the majority decisions of LSTM autoencoders trained separately on STE, STFT, and MFCC features. The observed performance improvement can be primarily attributed to the incorporation of a logistic regression classifier for anomaly discrimination, which proves to be more effective than conventional thresholding techniques in distinguishing between normal and anomalous sound instances.

V. CONCLUSION

This study investigates the application of autoencoder models in fusion with a logistic regression model for the development of a deep learning framework to detect anomalies in sound signal in industrial environments where early detection of equipment malfunctions is vital for ensuring operational efficiency. The proposed framework in this study comprises three distinct phases, namely: 1) Data Pre-processing phase 2) LSTM-autoencoder modeling phase which is exclusively trained with normal audio samples with an objective of reconstructing them as outputs with small errors, 3) Linear regression model (as an anomaly discriminator) designing phase which is trained with the reconstruction error vectors produced by the LSTM-autoencoder while fed with both normal and anomalous audio samples. Interestingly, the linear regression model adopted for distinguishing between normal and abnormal audio signals appears to be more effective than the popularly adopted thresholding strategy in autoencoder based anomaly detection framework. The proposed framework achieved a promising anomaly detection accuracy of approximately 97% when tested with "Fan Sound" dataset, which is derived from a subset of the MIMII Dataset (Malfunctioning Industrial Machine Investigation and Inspection).

Despite their effectiveness in modeling normal operational patterns and identifying deviations, autoencoder-based approaches for sound anomaly detection suffers from several limitations. The performance of autoencoders is heavily dependent on the quality and type of input features (e.g., MFCC, STFT, STE). Inadequate or noisy feature representations can impair the model's ability to reconstruct normal signals accurately, leading to false positives or negatives. Autoencoders trained solely on normal data may fail to generalize to unseen or slightly varying normal operating conditions, mistaking them for anomalies. This is particularly problematic in dynamic industrial environments with variable machine states or background noise.

While autoencoder-based systems have demonstrated substantial promise in unsupervised sound anomaly detection, Future research may focus on combining autoencoders with

advanced discriminative models such as support vector machines, random forests, or deep classifiers can enhance the ability to distinguish subtle anomalies from normal patterns, improving detection accuracy. Further, embedding attention layers within the autoencoder architecture may enable the model to focus on the most relevant portions of the input signal, thereby improving its sensitivity to subtle acoustic anomalies. The exploration of the above mentioned enhancements can push the boundaries of autoencoder-based sound anomaly detection, making it more reliable, interpretable, and deployable in diverse industrial contexts.

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